



How Does the **Internet** Actually Reach Your Phone?

FROM A SIMPLE TAP TO A GLOBAL NETWORK

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WHERE DID THIS VIDEO COME FROM?



The Internet is NOT Wi-Fi.

MISCONCEPTION

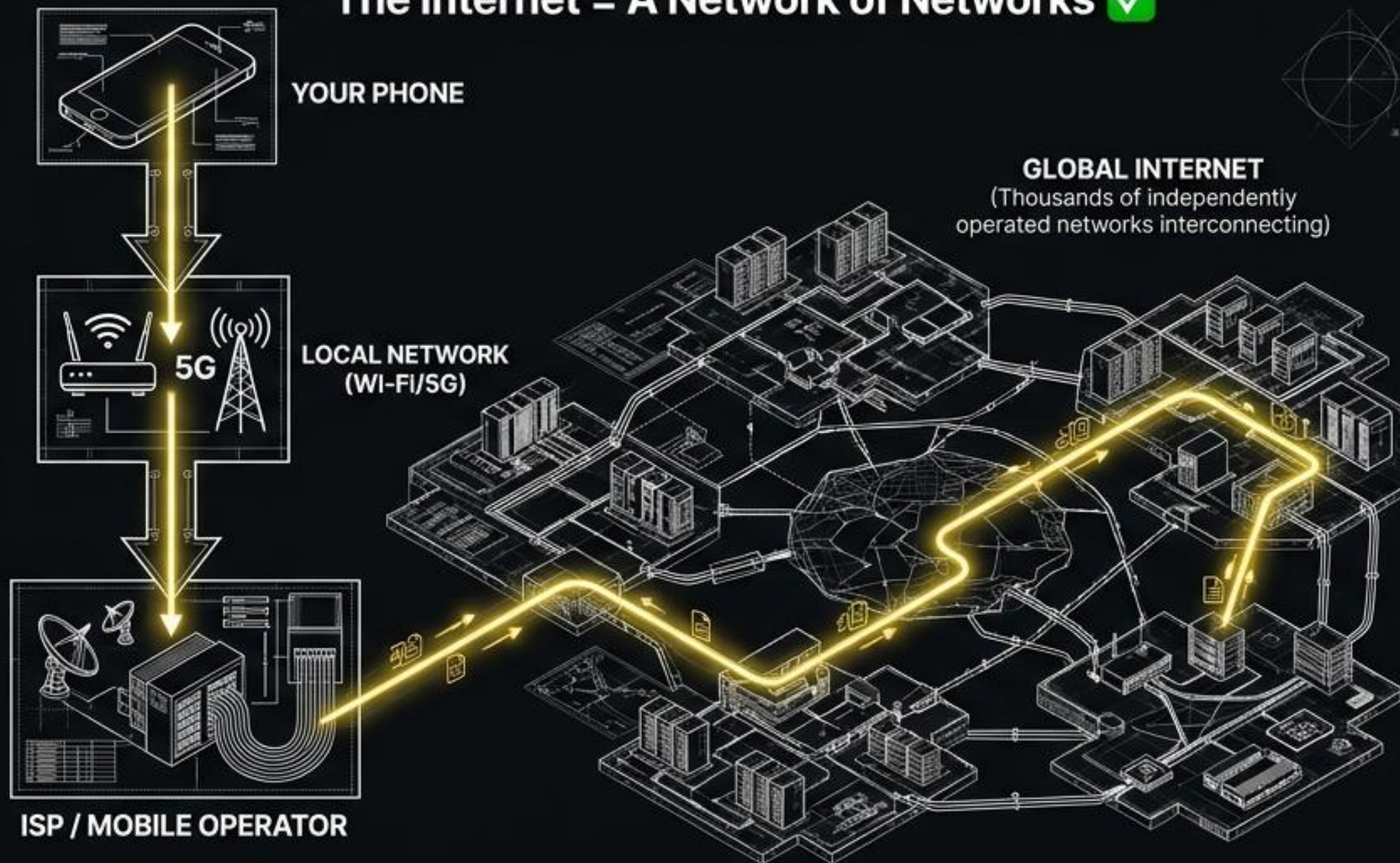
Wi-Fi = The Internet ❌



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REALITY

The Internet = A Network of Networks ✅



Your Phone's First Hop: Access

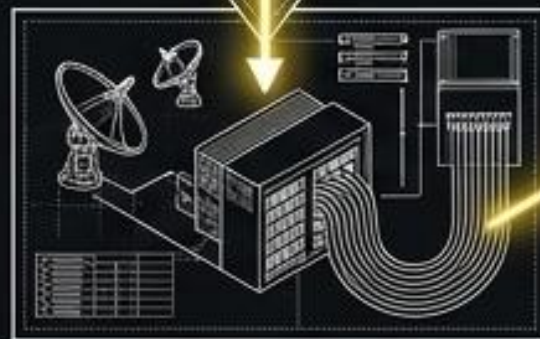
HOME ACCESS



(Radio Waves)



Wi-Fi Access
Point / Router



ISP Network

MOBILE ACCESS



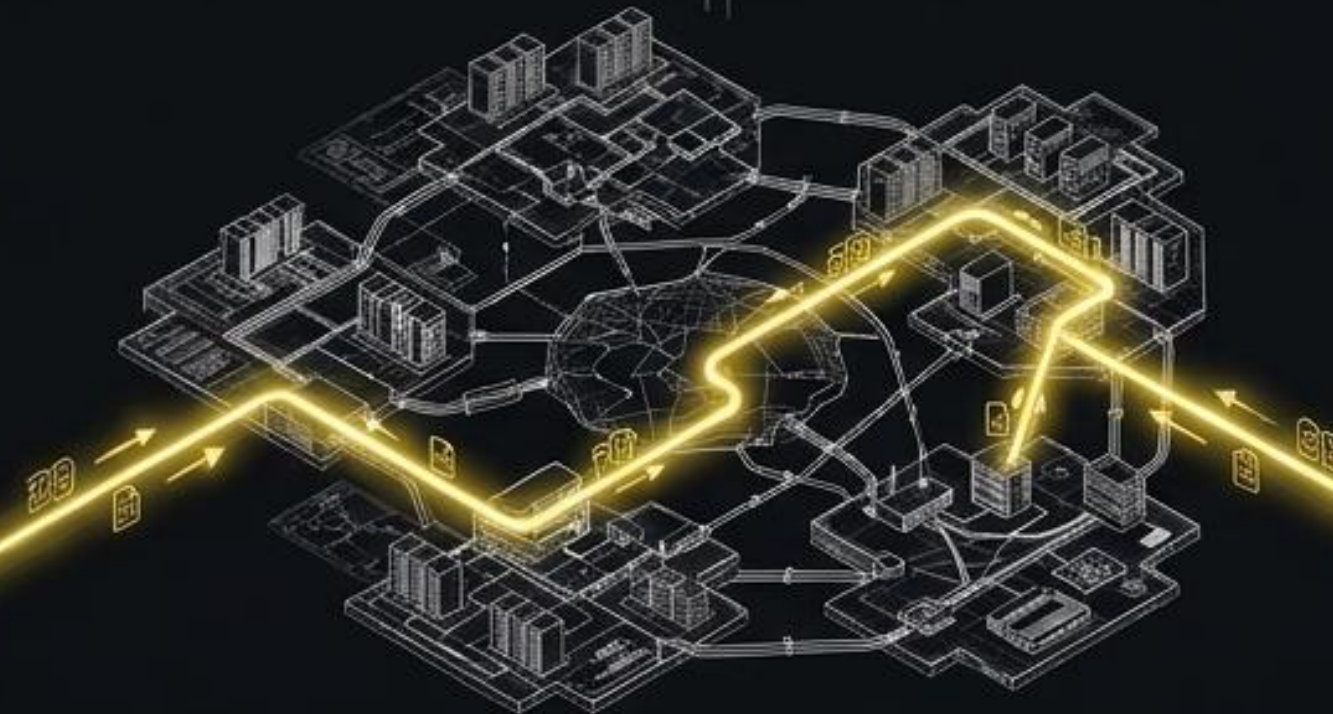
(Radio Waves)



Cell Site /
Base Station



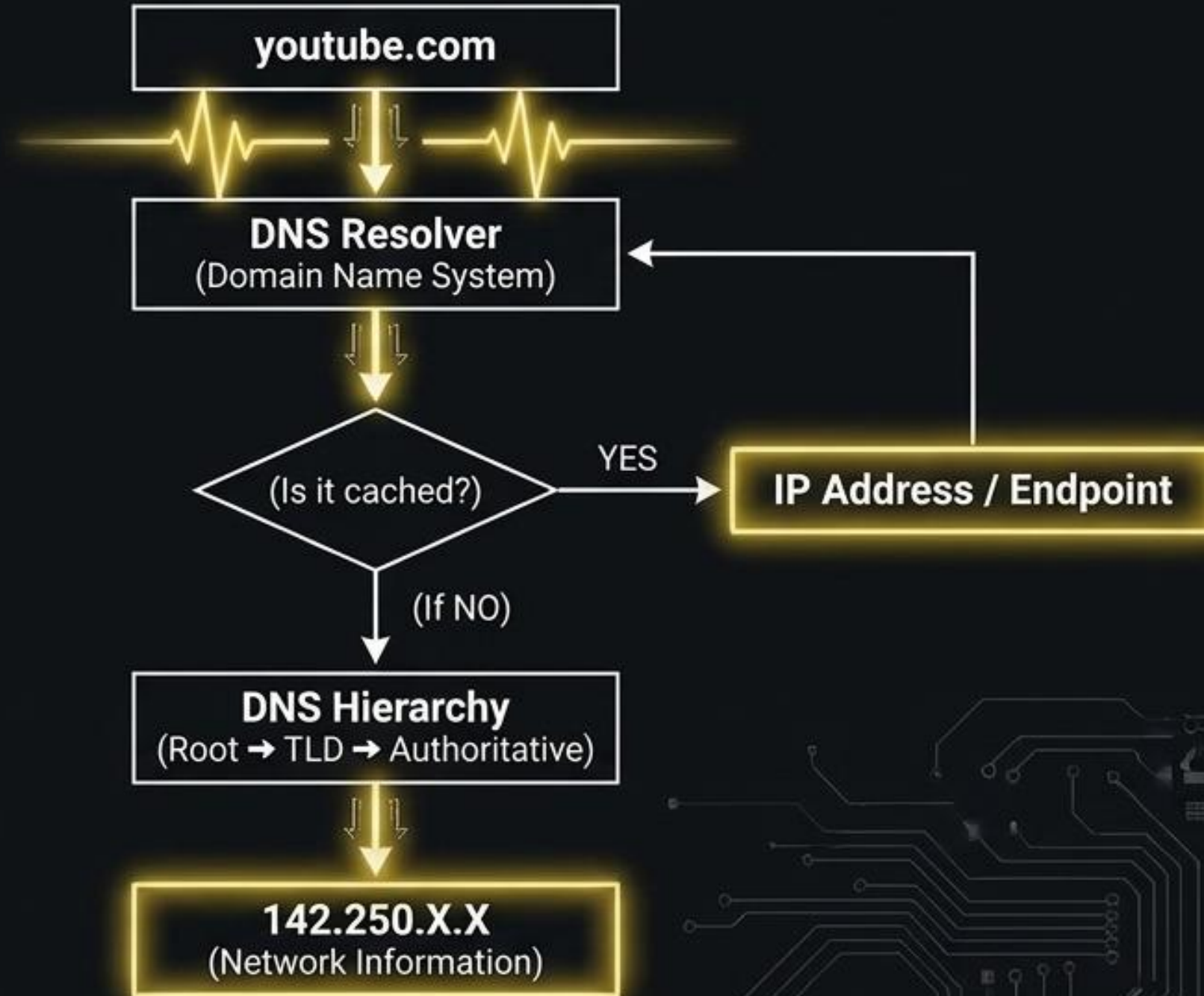
Mobile Operator Core



INTERNET

Before the Journey: Where is YouTube?

Translating Human Names to Network Addresses



Your Video Becomes Packets

Data travels in manageable units, not as one giant file.

THE REQUEST

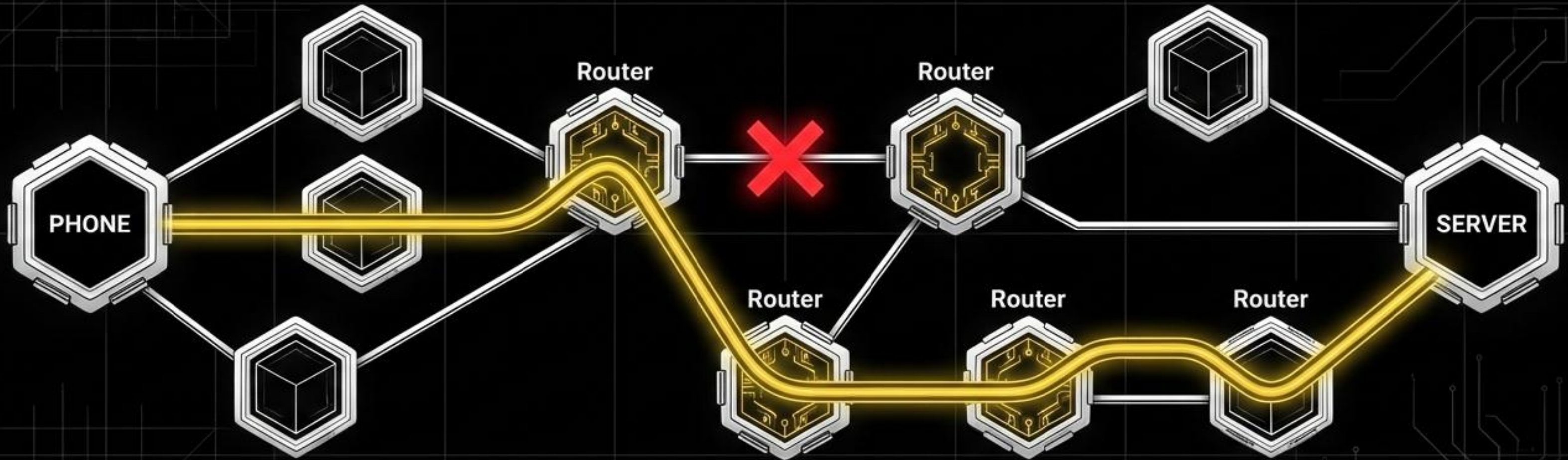


ANATOMY OF AN IP PACKET:

- ◀ Source IP Address (You) ◀ (Header)
- ◀ Destination IP Address (YouTube) ◀ (Header)
- ◀ Protocol Info (TCP/QUIC) ◀ (Header)
- ◀ Payload (The Video Data) ◀ (Data)

Routers: The Traffic Directors

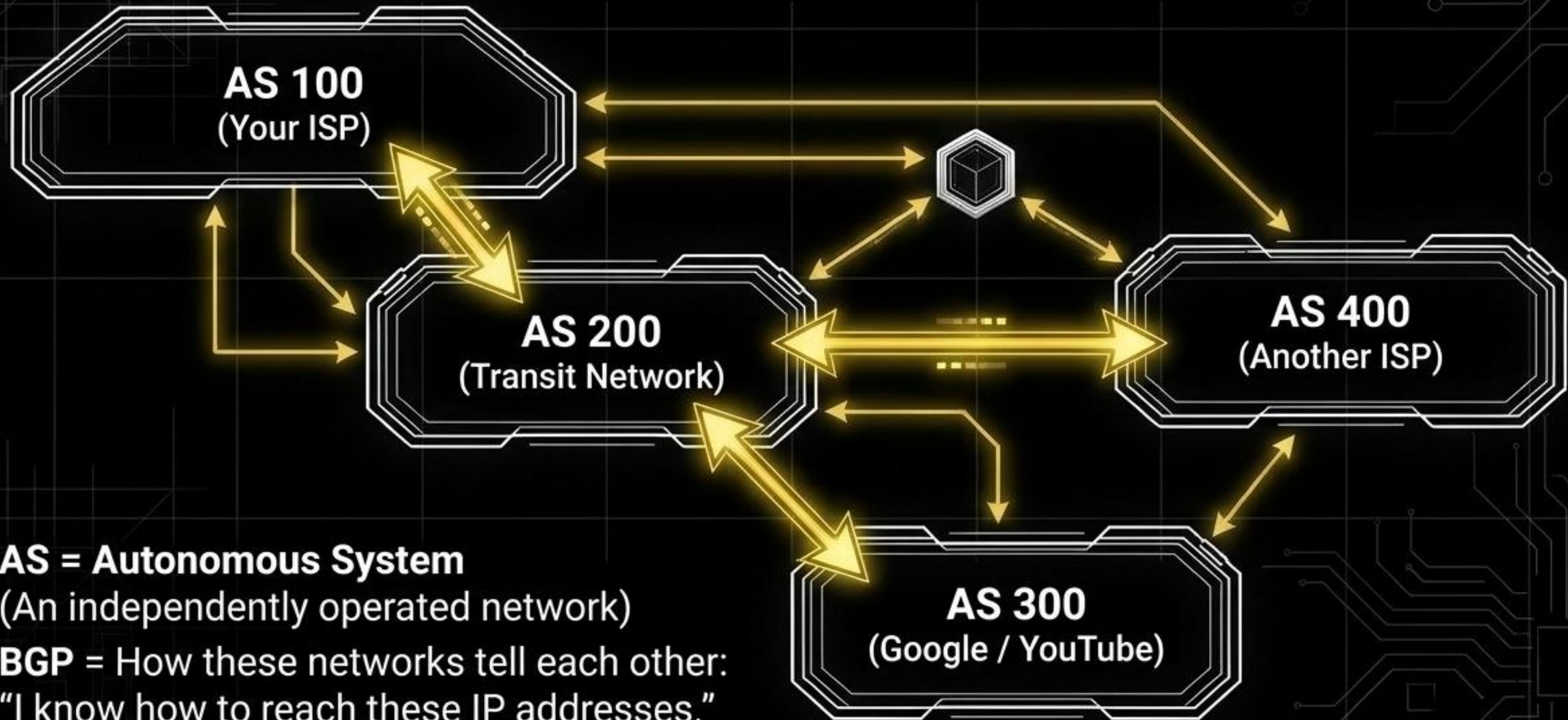
Deciding the “Next Hop” for every packet.



- * Routers read the destination IP.
- * No single router knows the whole journey.
- * **Paths can change dynamically.**

The Secret Language: BGP

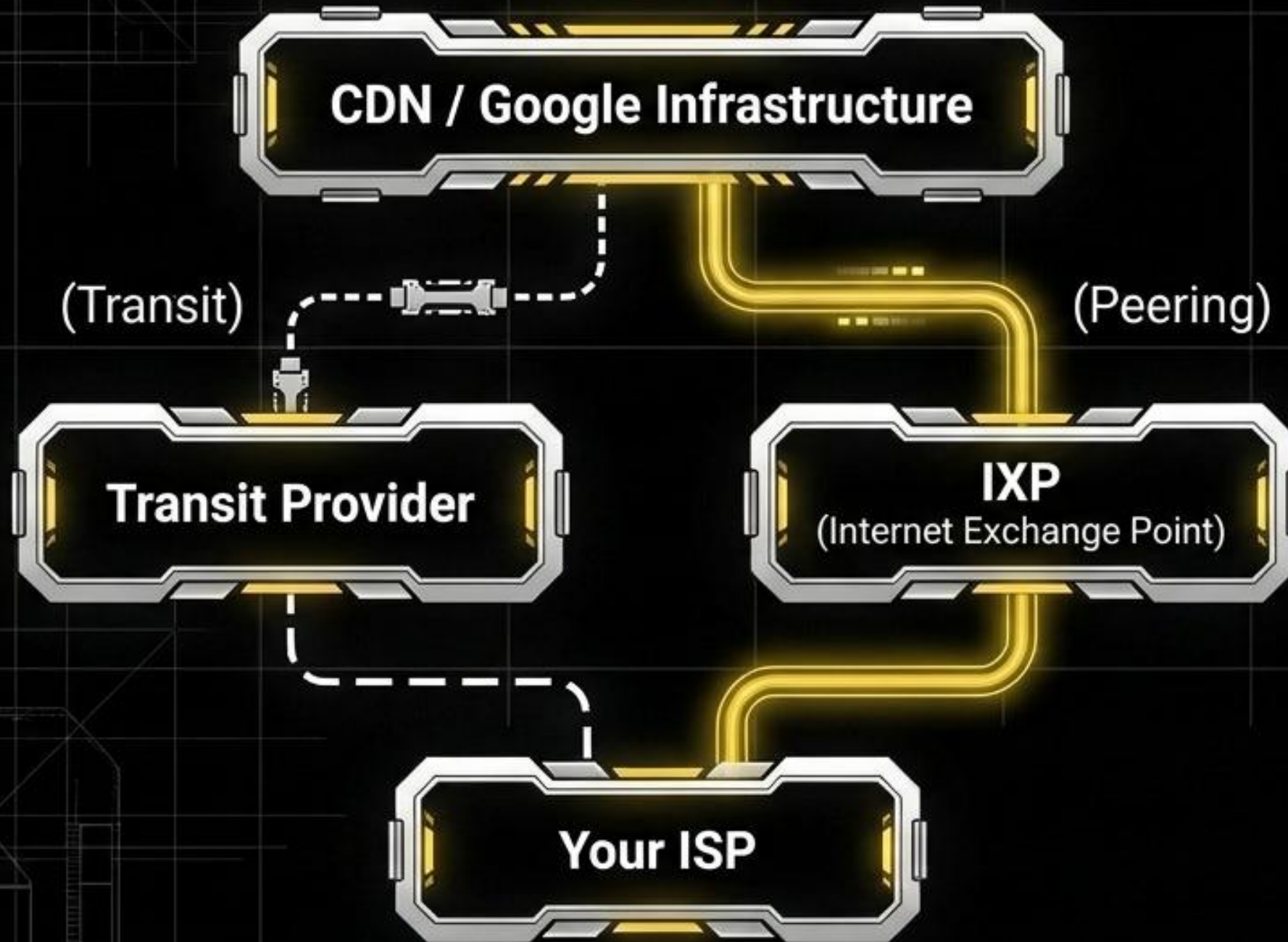
Border Gateway Protocol (BGP) connects independent networks.



- * **AS = Autonomous System**
(An independently operated network)
- * **BGP** = How these networks tell each other:
"I know how to reach these IP addresses."

Who Connects These Networks?

The architecture of global traffic exchange.



* **Transit:** Paying a larger network for global access.

* **Peering:** Networks connecting directly to trade traffic.

* **IXP:** Physical data centers where networks meet.

The Internet is Physical

There is no magical cloud.



THE "CLOUD" IS ACTUALLY:

- * **Fibre-Optic Cables**
(Optical signals in glass)
- * **Cellular / Wi-Fi**
(Radio waves)
- * **Data Centres**
(Servers, storage, cooling, power)
- * **Routers & Switches**
(Hardware)
- * **Submarine Cables**
(Deep-sea infrastructure)

**Satellites are complementary, not the primary global backbone.*

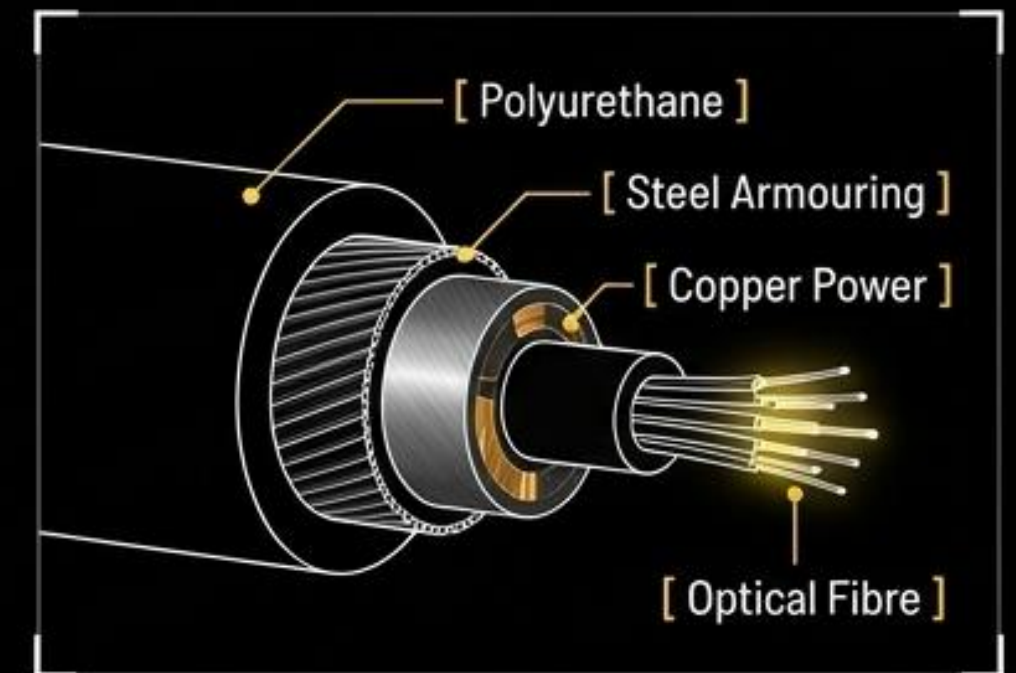
The Undersea Internet

How data crosses the oceans.

OVER 99% OF INTERNATIONAL DATA FLOWS VIA SUBMARINE CABLES.



INDIA → Cable Landing Station → **UNDERSEA CABLE** → Cable Landing Station → **DESTINATION**

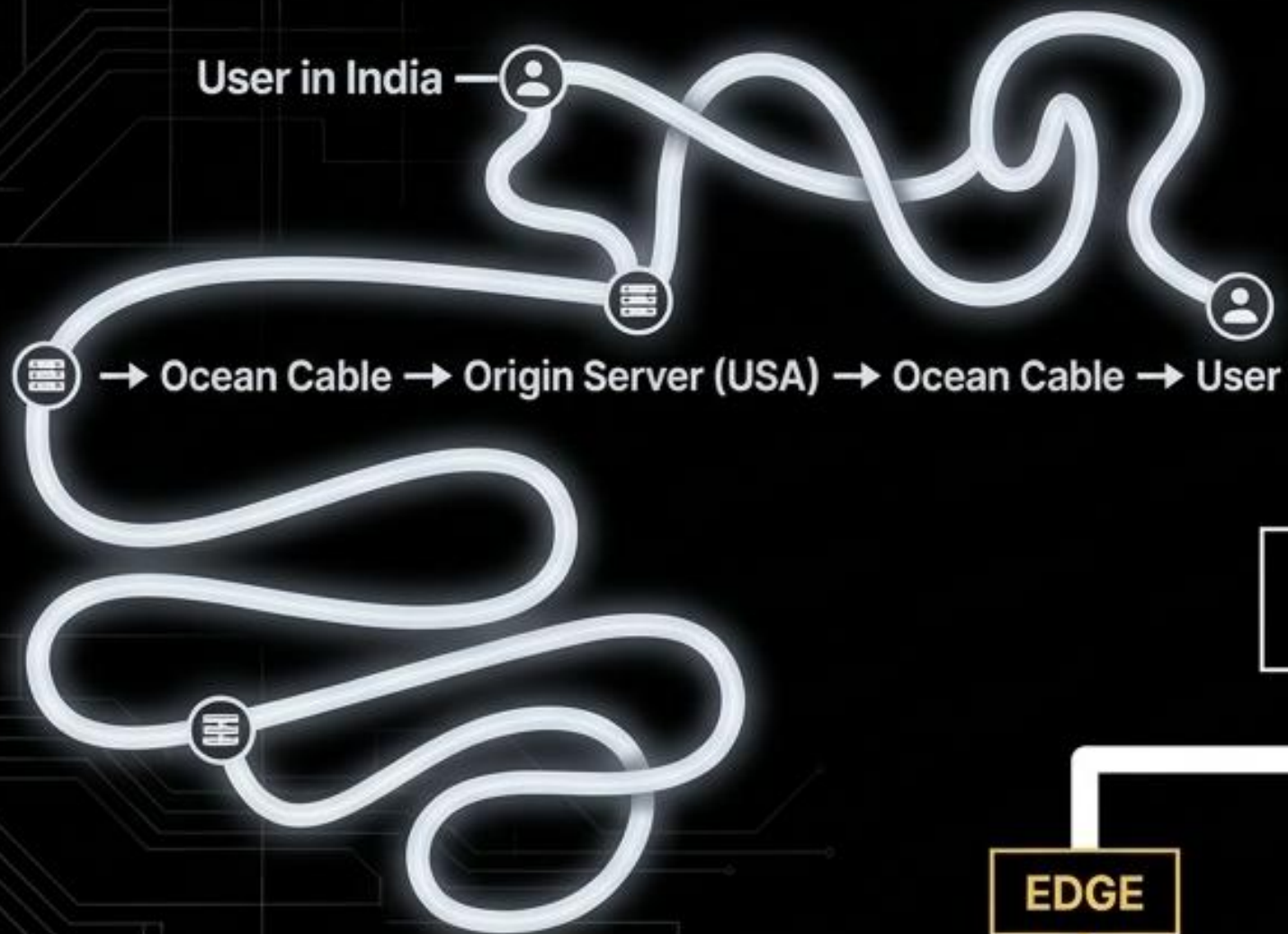


- * ~1.8 million kilometers of global cable
- * ~500 active cable systems

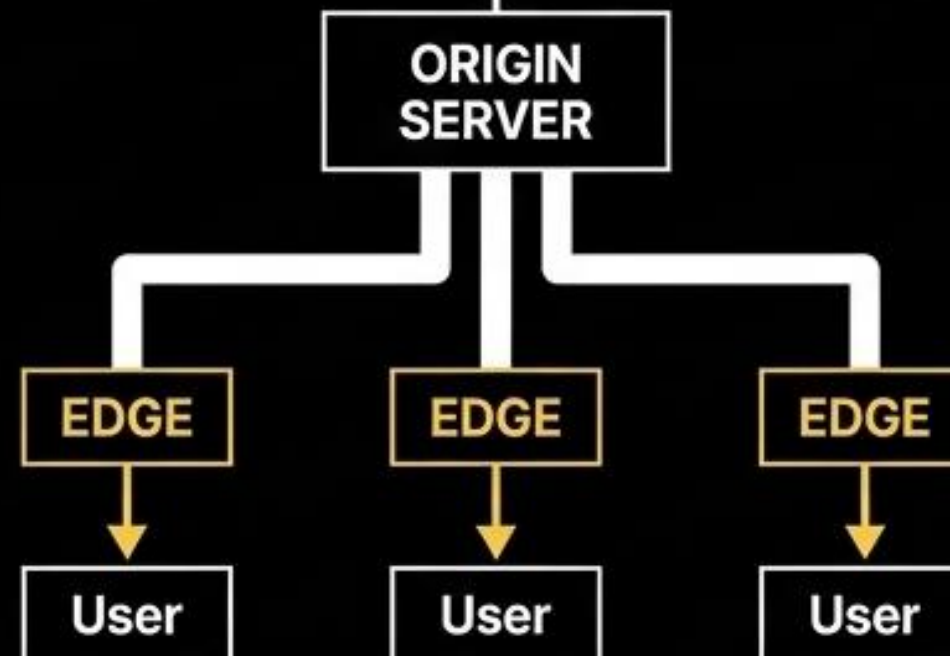
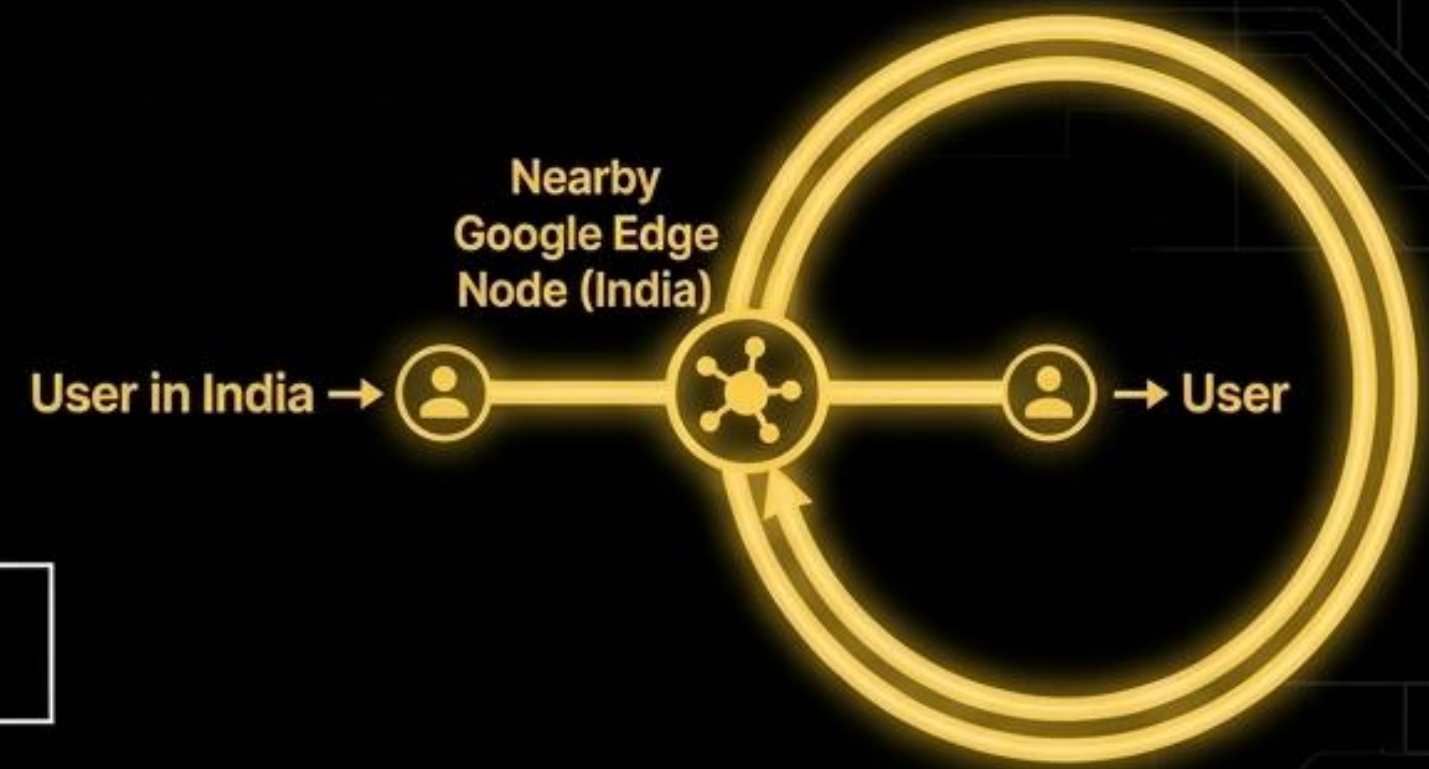
Wait... The Video May Not Cross the Ocean.

The power of Content Delivery Networks (CDNs).

WITHOUT CDN:

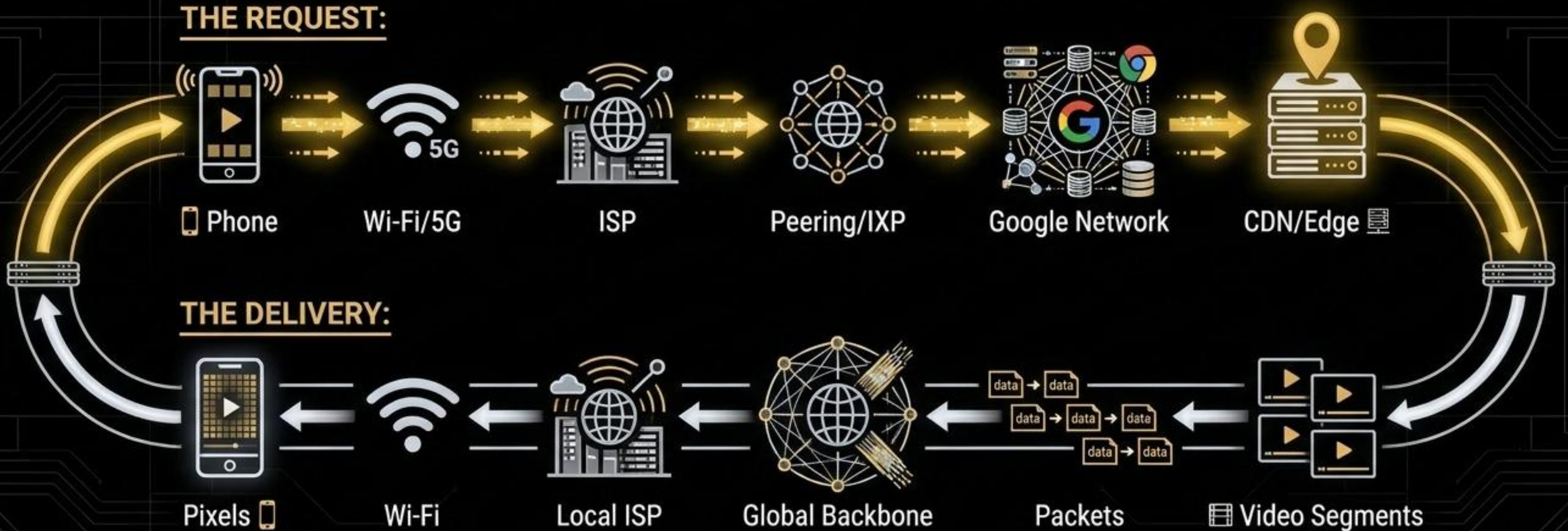


WITH CDN (Edge Caching):



Now Follow One YouTube Video

The Complete End-to-End Journey.



- * Connections established (TCP/QUIC/HTTP3).
- * Video requested dynamically.
- * Quality adapts in real-time.

Why Isn't The Internet Instant?

Understanding the physics of data delivery.

TOTAL LATENCY
(Travel Time)

≈

Propagation
(Distance/Speed of
Light)

+

Transmission
(Putting bits on
the wire)

+

Processing
(Routers reading
packets)

+

Queueing
(Waiting during
network congestion)



* **Bandwidth:** How wide is the pipe? (Capacity)

SPEED (Bandwidth) ≠ LATENCY

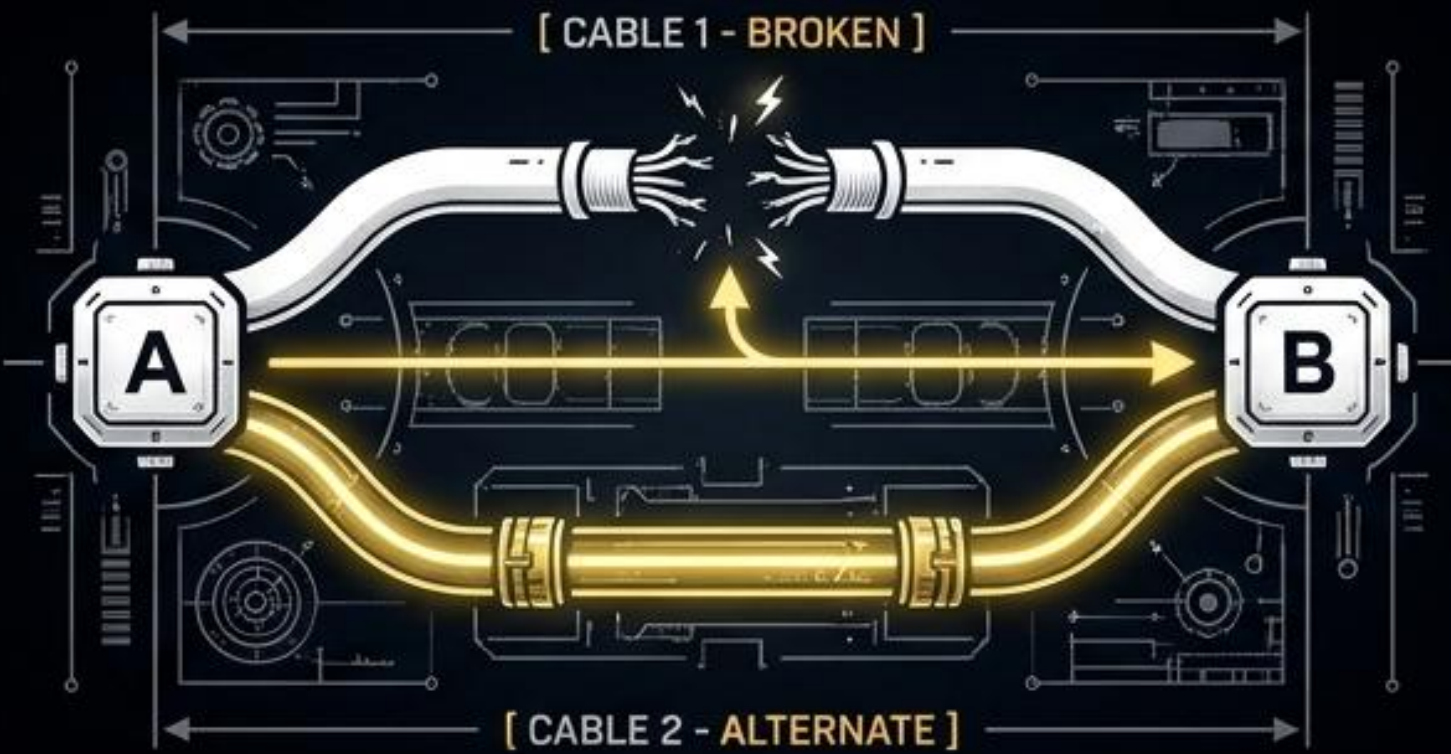


* **Latency:** How long does the trip take? (Delay)

What Happens When Things Go Wrong?

Infrastructure Resilience and Data Security

PHYSICAL FAILURE (Redundancy)



(~150-200 cable faults globally per year,
mostly human activity)

DATA SECURITY (Encryption)



- **HTTPS & TLS 1.3:** Encrypts the payload of your packets.
- **Note:** Observers can't see what video you are watching, but metadata (connecting, volume, IP) may still be visible.

The Internet is Everywhere...

...Because it is Nowhere.



TAP →



A planet-scale physical machine,
hidden behind a single tap on glass.



THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



PRESENTED BY
DAKSH KHANDELWAL
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KEEP LEARNING. KEEP BUILDING. KEEP GROWING.